

Multi-Hop Reasoning with RAG and Chain-of-Agents

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Abstract

Traditional Retrieval-Augmented Generation (RAG) pipelines struggle with multi-hop reasoning due to retrieval noise, missing intermediate facts, and difficulty in integrating dispersed information across documents. While the Chain-of-Agents (CoA) framework enhances reasoning through sequential information processing, it typically requires processing the entire corpus, which limits its scalability in real-world settings. To address these challenges, we propose a hybrid framework that builds upon RAG by incorporating a CoA-style reasoning module. Our system uses a semantic retriever to extract relevant evidence, which is then processed iteratively to distill and integrate critical information before generating a final answer. This structured approach reduces irrelevant content and improves factual consistency. Evaluated on the MultiHop-RAG dataset, our method demonstrates significant improvements in reasoning coherence and answer accuracy over baseline RAG systems.

1 Introduction

Large Language Models (LLMs) with extended context windows face challenges in retaining and utilizing mid-positioned information, a phenomenon known as "Lost in the Middle," which adversely impacts reasoning over lengthy inputs (Liu et al., 2023). Retrieval-Augmented Generation (RAG) addresses this by retrieving relevant external information rather than relying solely on internal memory (Lewis et al., 2020; Bowen Jin, 2024). However, traditional RAG methods remain insufficient for multi-hop question answering (QA) tasks (Tang and Yang, 2024; Zhuang et al., 2024).

Multi-hop QA demands integrating several retrieved facts to form accurate answers. For example, answering "What is the capital city of the country with the longest coastline in Africa?" necessitates first identifying Somalia, then retrieving its capital, Mogadishu. Traditional single-

pass RAG often misses intermediate facts because the information needed is spread across multiple documents, and the retrieved context may still include irrelevant content that limits reasoning performance. (Bowen Jin, 2024).

Conversely, the Chain-of-Agents (CoA) approach sequentially processes input, preserving contextual coherence and improving multi-hop reasoning but suffers from high computational costs (Zhang et al., 2024).

To overcome these limitations, we propose a hybrid architecture combining RAG with CoA. RAG retrieves a broad set of potentially relevant information, while CoA applies sequential processing to reduce the impact of irrelevant content and improve multi-hop reasoning. This approach ensures comprehensive evidence retrieval and structured reasoning, enhancing multi-hop QA performance, surpassing traditional long-context LLMs and standard RAG methods.

2 Related Work

2.1 Retrieval-Augmented Generation (RAG)

RAG (Lewis et al., 2020) improves long-context modeling by integrating retrieval with generative language models. It is widely used in several NLP tasks in open-domain QA, conversational agents, and personalized recommendation systems. For open domain QA, REAR (Wang et al., 2024) improves self-awareness of the relevance of the source, allowing the use of adaptive external knowledge. Similarly, (Chen et al., 2024) introduces a small encoder language model that efficiently encodes longer contexts while maintaining computational efficiency.

2.2 Multi-Hop QA in RAG

Multi-hop reasoning dataset like HotpotQA (Yang et al., 2018) is widely used in papers that proposed solutions for multi-hop queries but this dataset

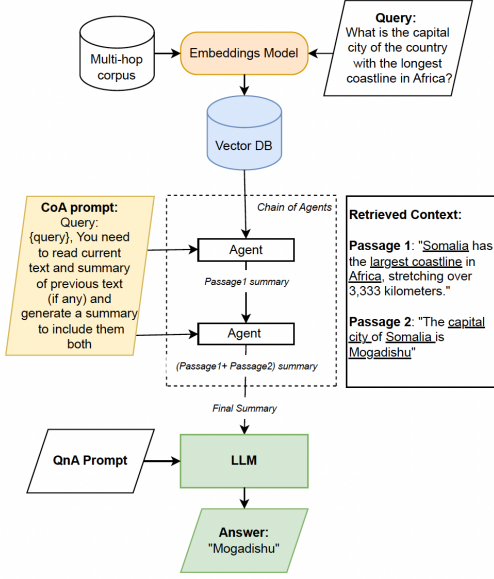


Figure 1: RAG with CoA Architecture

consists of queries that require at most two hops to reach the answer. MultiHop-RAG (Tang and Yang, 2024), on the other hand, provides a more diverse and challenging data that has documents with longer context length and queries that require more than two hops to deduce the answer. Hence, we chose to use this dataset as opposed to HotPotQA. Some potential solutions that demonstrate Multi-Hop question answering capabilities on HotPotQA include (Zhuang et al., 2024) where they iteratively refine queries to retrieve relevant information across multiple steps. Shi et al. (2024) provide a hierarchical retrieval approach in which a broad set of potentially relevant documents are retrieved and further narrowed down through subsequent retrieval stages.

However, these approaches lack the structured reasoning capabilities offered by the CoA framework, often relying on repeated retrieval calls to iteratively refine the query focus. In contrast, our proposed method utilizes a retriever to identify and narrow down the relevant context based on the query. This design improves reasoning while maintaining contextual relevance.

3 Problem Description

Traditional RAG retrieves relevant external information but falls short in multi-hop question answering, where synthesizing multiple interconnected facts is essential. Single-pass retrieval often misses critical intermediate evidence, resulting in retrieval

noise and factual inconsistencies (Zhuang et al., 2024). While the CoA approach enhances reasoning through sequential information processing, it assumes sequential processing of the entire corpus (Zhang et al., 2024), which is challenging for real-world applications with large-scale. We propose a system that employs RAG to retrieve relevant evidence from the entire document corpus, which is then passed to the CoA framework for sequential reasoning. This hybrid approach improves accuracy over standalone RAG and remains robust in handling large and diverse contexts, better reflecting real-world information settings. The retriever condenses long documents into concise factual segments, enabling more efficient processing by CoA’s LLM agents.

4 Methodology

4.1 Overview

As depicted in Figure 1, our system combines a Retrieval-Augmented Generation (RAG) approach with a CoA reasoning framework to address the challenges of multi-hop question answering. The approach aims to reduce retrieval noise, retain intermediate facts, and maintain coherence across long-context inputs. We evaluate our system using the MultiHop-RAG dataset, which contains complex queries requiring reasoning over multiple supporting documents. Each query is annotated with a gold-standard answer and its corresponding evidence passages, enabling assessment of retrieval and reasoning capabilities. For instance, answering the question “What is the capital of the country with the longest coastline in Africa?” involves identifying Somalia in one document and retrieving Mogadishu from another. Such queries serve to test whether the system can integrate dispersed facts to derive accurate answers. Our proposed solution is structured around four key stages: embedding and indexing the document corpus, retrieving passages semantically similar to the query, performing iterative reasoning through the CoA module, and generating the final answer using an LLM.

4.2 Initialization of Vector Database

The corpus from the MultiHop-RAG dataset, which comprises of 609 documents, is embedded using the pre-trained BAAI/bge-large-en-v1.5 model ¹, producing 1024-dimensional dense vectors. These

¹<https://huggingface.co/BAAI/bge-large-en-v1.5>

embeddings retain content and metadata such as title, source, and are stored in a Qdrant vector database, which enables fast and scalable similarity-based retrieval.

4.3 Initial Retrieval Process

Each query is embedded with the same model as used for corpus embeddings. We retrieve the top- k chunks from the vector database that are most semantically similar to the embedded query. To further improve relevance, the retrieved passages are reranked based on cosine similarity scores. Although this initial retrieval typically captures relevant information, it may also include irrelevant or noisy passages. To mitigate the impact of such noise, the retrieved passages are forwarded to the CoA module, which conducts structured, iterative reasoning to distill the most pertinent information.

4.4 Chain of Agents (CoA) Module

The CoA module takes the retrieved passages and processes them sequentially using a fixed LLM. In the first iteration, the LLM is prompted with the first chunk and the original query as the task prompt. In each subsequent iteration, it receives the current chunk and the summary from the previous step as context, along with the same task prompt. The LLM serves as the worker agent in each step, tasked with extracting and summarizing query-relevant information from the current chunk. This iterative summarization continues across all chunks, building up a coherent reasoning trail. Once all chunks have been processed, a manager agent, also implemented using the same LLM, is provided with the aggregated summary and a task-specific QnA prompt to generate the final answer. This structured reasoning enables the model to preserve and integrate intermediate facts while reducing the influence of hard negatives, thereby enhancing performance on multi-hop questions.

5 Experiments and Results

5.1 Experimental Setup

Dataset: The MultiHop-RAG dataset comprises 2,556 multi-hop queries based on English news articles. Each query is supported by evidence from 2–4 documents and falls into one of four categories: inference, comparison, temporal, or null (unanswerable) queries.

Baselines: For comparative analysis we employ standard long-context language models with

out explicit retrieval, namely **LLaMA-3.3**² and **DeepSeek-R1**³. Next, we include a **Traditional RAG** pipeline (Lewis et al., 2020) that uses Qdrant as the vector database, BAAI/bge-large-en-v1.5 embeddings for retrieval and LLaMA-3.3 for response generation. Furthermore, we evaluate the standalone **Chain of Agents (CoA)** method (Zhang et al., 2024). We test both LLaMA-3.3 CoA and DeepSeek-R1 CoA, with and without multi-turn instructions. In some settings, workers are asked to generate concise 2 to 3 line responses to improve interpretability.

Evaluation Protocols: We compute ROUGE (ROUGE-1, ROUGE-2, and ROUGE-L) (Lin, 2004), METEOR (Banerjee and Lavie, 2005), and BERTScore (Precision, Recall, and F1) (Zhang et al., 2020) to evaluate the quality of the responses generated. For assessing the retriever’s performance, we use Mean Reciprocal Rank (MRR) (Craswell, 2009).

5.2 Results and Analysis

Table 1 shows comparison between the baseline models and our proposed method. LLaMA-3.3 with $k = 10$ shows strong performance, outperforming standalone LLaMA 3.3 across all metrics with notable improvements in METEOR (+7.7%) and BERTScore-F1 (+21.8%). While standalone DeepSeek-R1 shows better ROUGE scores, our model performs better on semantically oriented metrics like BERTScore-F1 (+3.6%) and METEOR, suggesting that our outputs are more fluent and meaningfully aligned with the references, even if they diverge slightly at the surface level. Against CoA LLaMA-3.3, our model shows comparable performance, with a minimal gap of just 0.93% in BERTScore-F1. The improved CoA variant (LLaMA-3.3*) achieves higher scores, including 7.8% better METEOR and 4.7% higher BERTScore-F1. The significant performance improvements observed in CoA and DeepSeek-R1 baselines are probably due to explicitly provided evidence lists during the evaluation due to their lack of native long-context support, which probably contributed to their high scores.

Compared to the traditional RAG system, our

²LLaMA-3.3 refers to LLaMA-3.3-70B-Instruct-Turbo:<https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct>

³DeepSeek-R1 refers to DeepSeek-R1-Distill-Llama-70B:<https://huggingface.co/deepseek-ai/deepseek-11m-7b-chat>

Group	Model	Rouge-1	Rouge-2	Rouge-L	METEOR	BERT-P	BERT-R	BERT-F1
Baselines	LLaMA-3.3	0.6191	0.2217	0.6194	0.3624	0.657	0.688	0.668
	DeepSeek-R1	0.7170	0.1667	0.7160	0.4331	0.9053	0.9266	0.9140
	RAG	0.4500	0.1900	0.4500	0.3000	0.800	0.810	0.800
	CoA (LLaMA-3.3)	0.6754	0.2405	0.6753	0.4180	0.8943	0.8874	0.8906
	CoA (LLaMA-3.3*)	0.7259	0.2445	0.7265	0.4449	0.9196	0.9114	0.9151
	CoA (DeepSeek-R1)	0.6029	0.1862	0.6022	0.3832	0.8532	0.8737	0.8621
Proposed Method: RAG + CoA								
LLM Choice	LLaMA-3.3 (k=10)	0.6266	0.2214	0.6262	0.3900	0.8769	0.8724	0.8744
	DeepSeek-R1 (k=10)	0.6029	0.1862	0.6022	0.3832	0.8532	0.8737	0.8621
Varying Top-k	LLaMA-3.3 (k=15)	0.6273	0.2222	0.6268	0.3909	0.8755	0.8710	0.8730
Re-Retrieval Between Agents	LLaMA-3.3 (k=10, reretrieval k=4)	0.6279	0.2223	0.6271	0.3910	0.8757	0.8711	0.8731
	DeepSeek-R1 (k=10, reretrieval k=4)	0.6042	0.1876	0.6052	0.3842	0.8520	0.8724	0.8608
Reranking	LLaMA-3.3 (k=10, reranking)	0.6256	0.2044	0.6250	0.3823	0.8746	0.8699	0.8719

Table 1: Evaluation on the MultiHop-RAG dataset for baselines and proposed model variants. Here, Llama-3.3 refers to *LLaMA-3.3-70B-Instruct-Turbo* and DeepSeek-R1 refers to *DeepSeek-R1-Distill-Llama-70B-free*. The "*" denotes experiment where worker agents in CoA was prompted to generate brief 2–3 line responses. Variants are grouped based on design factors such as LLM, retrieval depth, and reranking.

Hits@10	Hits@4	MAP@10	MRR@10
0.7397	0.5463	0.1787	0.3937

Table 2: Overall retrieval performance of traditional RAG.

method achieves significant gains: +39.2% in Rouge-1 and Rouge-L, +16.3% in Rouge-2, +30% in METEOR, and +9.6% in BERTScore-F1. These results highlight the strength of our multi-hop CoA framework with dynamic retrieval, enabling it to generate responses that are more context-aware, informative, and coherent.

5.3 Ablation Study

We perform four ablation studies to assess the impact of key design choices in our system. **LLM Choice:** We check the impact of different base language models, comparing LLaMA-3.3 and DeepSeek-R1 (both with $k=10$). We see that Deepseek-R1 ($k=10$) achieves 2.4% lower Rouge metrics, 0.7% lower METEOR, and 1.2% lower BERTScore-F1, indicating that LLaMA 3.3 ($k=10$) is a better choice and more effective in producing multi-hop responses. **Varying Top-K:** we vary the initial Top-k evidence retrieval (k) using LLaMA-3.3. Comparing $k = 10$ and $k = 15$, we observe marginal improvement across all metrics like +0.13% Rouge-1, +0.23% METEOR, with BERTScore-F1 slightly declining by 0.14%. These small differences suggest that retrieving more documents offers limited benefit. **Re-Retrieval Between Agents:** We experimented with re-retrieval between worker agents, where instead of relying solely on the initial top- $k = 10$ passages, every alternate agent performs a secondary retrieval ($k = 4$) based on the current summary and question to fetch more targeted context for the next reasoning step. We per-

formed this with Llama3.3 and DeepSeek-R1, both showing negligible improvements. **ReRanking:** re-ranking list of evidence documents based on cosine similarity between document and query embeddings (with $k = 10$) led to the best gains: +0.5% in Rouge-1, +0.23% in METEOR, and +0.45% in BERTScore-F1, confirming that prioritizing semantically relevant context early on improves generation quality.

Retrieval performance is evaluated by checking whether the gold evidence passages appear among the top-k retrieved results. As shown in Table 2, traditional RAG shows effective retrieval, with Hits@10 of 0.7397 and Hits@4 of 0.5463, meaning relevant documents are often found within the top results. The MAP@10 of 0.1787 and MRR@10 of 0.3937 indicate that relevant passages are not only retrieved but tend to appear early in the ranking.

6 Conclusion

In conclusion, our multi-hop, context-aware approach outperforms both traditional RAG and standalone models. By combining retrieval with agent-based reasoning, it handles multi-hop queries effectively and proves to be both robust and practical.

7 Future Work

Future work can be improving retrieval recall, testing on diverse datasets such as HotpotQA, and exploring LLMs with larger context windows. Additionally, we aim to refine the CoA by updating the query during re-retrieval to identify what information is still missing in the retrieved evidence. Finally, replacing basic cosine re-ranking with advanced methods like cross-encoder models could further improve generation quality.

8 Individual Contributions

Santoshi Vishnu Manasa Bellamkonda Evaluated the LLaMA model on the MultiHop-RAG dataset using input prompts constructed by combining the query with corresponding evidence documents as context, and analyzed the quality of the model's responses using evaluation metrics. Filtered out queries with input lengths exceeding the token limit supported by the free API, resulting in final set of queries used for evaluation. Conducted re-retrieval experiments within our proposed architecture using the LLaMA model to assess its impact on reasoning performance.

Rajas Chitale Implemented CoA architecture and evaluated the MultiHop-RAG dataset on it with Meta-LLaMA-3-70B-Instruct and DeepSeek-R1-Distill-Llama-70B as backbone. Developed an organized strategy for studying and presenting related work. Implemented and evaluated the hybrid system utilizing RAG and CoA.

Gauri Naik Conducted experiments to evaluate the performance of the standalone DeepSeek-R1-Distill-LLaMA model and Re-Retrieval Between Agents on the DeepSeek-R1-Distill-LLaMA model. Built a prompt generation pipeline to retrieve and format evidence passages from the corpus based on the evidence titles provided for each query. Computed evaluation metrics for each generated response.

Shivaraj Bhanji Designed and implemented the RAG pipeline for MultiHop-RAG dataset, integrating LlamaIndex for embedding generation and storing embeddings in Qdrant DB for efficient vector search. Used the LLaMA-3.3-70B-Instruct-Turbo model for inference. Evaluated retrieval performance and model generation, as shown in [Table 2], to benchmark performance. Researched about the available multi-hop datasets and helped with project presentation.

Aditya Sharma Laid the Roadmap for RAG evaluations, identifying models, various techniques and frameworks. Researched embedding models on hugging face, implemented Python code to enable use of LLM models, setup LLM model inference on local machine for testing with small parameter models. Incorporated an embedding-based re-ranking step using cosine similarity between the query and retrieved contexts to determine the evi-

dence order passed to the CoA framework for multi-hop reasoning.

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